Design and PLECS Simulation of a High-Power-Quality Single-Phase Rectifier

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Abstract

This report presents the design and simulation of a single-phase, 230 V / 50 Hz, 4 kW rectifier with a regulated 400 V DC output that must satisfy a strict set of power-quality targets: power factor above 0.99, source-current THD below 3 %, THD of the voltage at the point of common coupling (PCC) below 2 %, and a DC-side form factor close to one. A passive diode bridge with a bulk capacitor cannot meet these limits, so an active power-factor-correction (PFC) stage is used. A two-phase interleaved boost converter operating at 100 kHz was selected; because at 4 kW, it halves the current handled by each inductor and switch, cancels part of the switching ripple at the input, and reduces the RMS current in the output capacitor. The power stage (2 × 250 µH, 3400 µF, damped EMI filter) was dimensioned analytically and a cascaded control structure was built: a slow voltage loop that outputs a power reference, a reference generator that shapes the inductor-current reference in proportion to the rectified line voltage, and two fast, independent inductor-current loops with duty-cycle feed-forward. The complete system was simulated in PLECS. The simulation reports PF ≈ 0.99996, source-current THD = 0.68 %, PCC voltage THD = 0.42 % and form factor = 1.0000 at an output power of 4.00 kW, satisfying every specification.

Note: To be honest, I know that something this complex wasn't expected, but I wanted to push myself a bit further; consequently, I have no idea how it might be evaluated.

1. Problem Statement and Specifications

The assignment specifications are collected in Table 1.

Table 1. Assignment specifications.

Quantity	Specification	Remark
Source voltage	230 V RMS, 50 Hz	Vpk = 325.3 V
Source inductance	700 µH (inductive)	Lg, X = 0.22 Ω at 50 Hz
Output power	4000 W	DC side
Power factor	> 0.99	AC side
THD of source current	<3 %	AC side
THD of voltage at PCC	<2 %	PCC = rectifier input term. behind Lg
Form factor of output voltage	close to 1	FF = Vo,rms / Vo,avg

The 700 µH source inductance is not a design variable. It is the grid impedance seen by the rectifier, and it is what makes the PCC voltage sensitive to the harmonics of the drawn current: any distorted current produces a distorted voltage drop across it. The two AC-side THD limits are therefore linked, and both are met only if the input current is almost perfectly sinusoidal.

2. Choice of Topology

2.1 Why a passive rectifier is not enough

A diode bridge feeding a capacitor draws current in short pulses near the voltage peaks. Its power factor is typically below 0.7 and its current THD is far above the 3 % target, so the specification cannot be met with a passive rectifier, even with a large line inductance. The practical way to reach PF> 0.99 and THD <3 % at 4 kW is an active PFC stage that forces the input current to follow the rectified line voltage. Consequently, I assumed that an inductor with a very large value would not be very practical in a real-world scenario.

2.2 Why a boost PFC

The boost converter is a common used PFC topology. The inductor sits at the input, so the input current is continuous and can be shaped by average-current-mode control.

2.3 Why interleaving

At 4 kW and 230 V the input current is 17.4 A RMS and 24.6 A peak. A single boost stage would have to carry all of it in one inductor and one switch. Because this is a practical, high-power application, a two-phase interleaved boost was chosen. Two identical boost stages operate in parallel with their carriers shifted by 180°, and the input current is the sum of the two inductor currents. This gives the following benefits, all documented in the reference material [1], [2]:

- Each inductor and switch carries 12.3 A peak instead of 24.6 A, which lowers conduction loss and spreads the heat over two devices.
- The two triangular inductor ripples are out of phase and partly cancel at the input (Section 3.4), which shrinks the EMI filter; the ripple seen at the input is at twice the switching frequency.
- The two diode currents interleave, so the high-frequency capacitor current is roughly halved (Section 3.5).
- For equal inductance the total stored inductor energy is half of the single-stage value [2], which allows smaller cores.

The price is a second inductor, switch and diode, and the need to keep the two phases balanced. In this design each phase has its own current loop with the same reference, so the currents are balanced by construction (Section 4).

2.4 Reference designs studied

Two documents were used as references. The first is the ROHM 3.6 kW interleaved boost PFC reference design [1], a real 180–265 V AC / 390 V DC board with two boost phases at 65 kHz, 680 µH inductors, four 680 µF bulk capacitors and a digital controller with a voltage loop, a current loop and a current-balancing loop. The second is the Texas Instruments seminar paper by M. O'Loughlin [2], which derives the ripple-cancellation factor K(D), the output-capacitor RMS current and the inductor-energy reduction for two to four phases and validates them on a 350 W prototype. Their design procedure was followed here and re-scaled to the 4 kW / 400 V / 100 kHz operating point of this assignment.

3. Power-Stage Design

The converter of Fig. 1 consists of the 700 µH source, a damped input filter, a diode bridge, two boost phases (L_{b1}–FET1–D5 and L_{b2}–FET2–D6) and the bulk capacitor feeding a resistive load. The parameters used in the PLECS model are derived.

3.1 Operating point

$$V_{pk} = \sqrt{2}\, V_{rms} = 325.3\ V \tag{1}$$

$$R_{load} = \frac{{V_o}^2}{P} = \frac{400^2}{4000} = 40\ \Omega,\ \ I_o = \frac{P}{V_o} = 10\ A \tag{2}$$

For a unity-power-factor, lossless converter the input current is:

$$I_{in,rms} = \frac{P}{V_{rms}} = 17.39\ A,\ \ I_{in,pk} = \sqrt{2}\, I_{in,rms} = 24.6\ A \tag{3}$$

and, since it is shared equally by the two phases, the peak current reference of each inductor is:

$$I_{ph,pk} = \frac{I_{in,pk}}{2} = \frac{P}{V_{pk}} = 12.3\ A,\ \ I_{ph,rms} = \frac{I_{ph,pk}}{\sqrt{2}} = 8.70\ A \tag{4}$$

3.2 Duty cycle

In continuous conduction mode the boost conversion ratio gives the instantaneous duty cycle of each switch. With the rectified line voltage $v_{in}(\theta) = V_{pk}|\sin\theta|$,

$$D(\theta) = 1 - \frac{v_{in}(\theta)}{V_o} = 1 - \frac{V_{pk}\,|sin\ \theta|}{V_o} \tag{5}$$

The duty cycle is 1 at the zero crossings and falls to its minimum of $D_{min} = 1 - 325.3/400 = 0.187$ at the line peak (Fig. 1, left). It passes through D = 0.5 when v_{in} = 200 V, which is the point of perfect ripple cancellation. The duty-cycle command is limited to 0.95 in the controller so that the switch always has a minimum off-time.

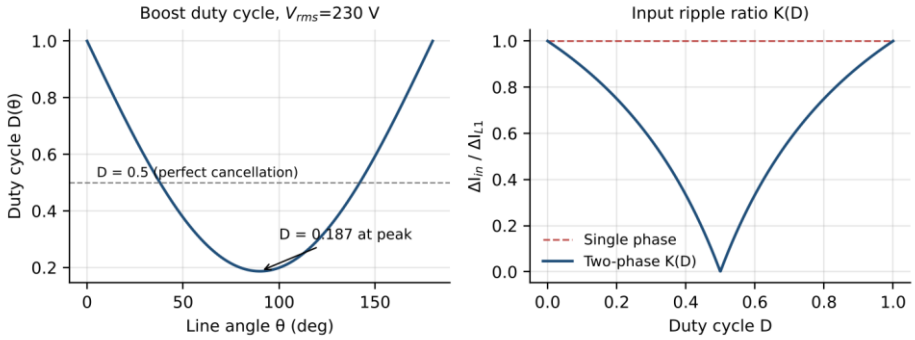


Figure 1. Duty cycle over a half line cycle (left) and the input-to-inductor ripple ratio

3.3 Boost inductors

The peak-to-peak inductor ripple at line voltage v_{in} is:

$$\Delta I_L(v_{in}) = \frac{v_{in}\left(1 - \frac{v_{in}}{V_o}\right)}{L_b\, f_{sw}} \tag{6}$$

This is largest at $v_{in} = V_o/2$ (D = 0.5), where:

$$\Delta I_{L,max} = \frac{V_o}{4\, L_b\, f_{sw}} \ \Rightarrow\ L_b = \frac{V_o}{4\, \Delta I_{L,max}\, f_{sw}} \tag{7}$$

A worst-case ripple of 4 A peak-to-peak, i.e. about 33 % of the 12.3 A per-phase peak, was accepted. This is larger than the 20 % often used for a single-stage design, because interleaving removes much of it at the input and because a higher switching frequency (100 kHz instead of 65 kHz in [1]) keeps the inductance small. The result is:

$$L_b = \frac{400}{4 \cdot 4 \cdot 100\,000} = 250\ \mu H \tag{8}$$

At the line peak the ripple is 2.43 A, so each inductor reaches a maximum current of about 12.3 + 1.2 ≈ 13.5 A. Note that the same worst case D = 0.5 is also the point where the interleaved input ripple is zero. Near the zero crossings the converter enters discontinuous conduction for a short time because the average inductor current becomes smaller than half the ripple; the measured THD of Section 6 shows that this has little effect on the current quality.

3.4 Ripple cancellation at the input

The ratio between the ripple of the summed input current and that of one inductor follows from the superposition of the two phase-shifted triangular currents [2]:

$$K(D) = \frac{\Delta I_{in}}{\Delta I_{L1}} = \frac{1-2D}{1-D},\ D \le 0.5; \quad K(D) = \frac{2D-1}{D},\ D > 0.5 \tag{9}$$

K is zero at D = 0.5 and rises towards 1 at both ends (Fig. 1, right). Over the line cycle the input ripple is $\Delta_{in}(\theta) = K(D(\theta)) \cdot \Delta I_L(\theta)$; its maximum is 2.0 A at v_{in} = 300 V (D = 0.25), compared with 4.0 A for a single stage with the same inductance (Fig. 2). At the line peak, K = 0.77, so the 2.43 A inductor ripple becomes 1.87 A at the input. Besides the amplitude, the ripple frequency seen by the source is 200 kHz, which is easier to filter.

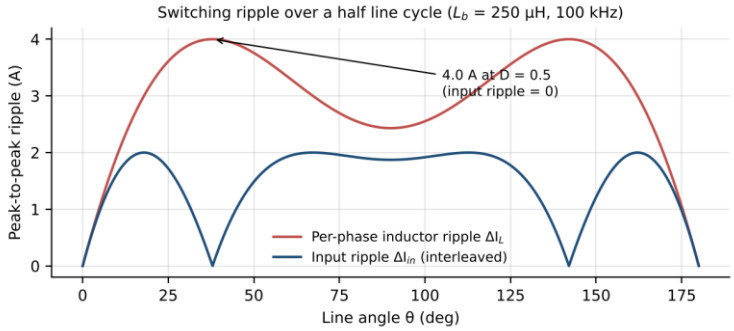


Figure 2. Inductor and input ripple over a half line cycle for Lb = 250 µH and fsw = 100 kHz

3.5 Output capacitor

Two criteria were used. The first is the hold-up requirement of the reference design [1]: the bus must ride through half a line cycle (t = 10 ms) without falling below V_{min} = 350 V,

$$C_{out} \ge \frac{2\,P\,t}{V_o^{\,2} - V_{min}^{\,2}} = \frac{2 \cdot 4000 \cdot 0.01}{400^2 - 350^2} = 2133\ \mu F \tag{10}$$

The second is the 100Hz voltage ripple caused by the pulsating single-phase power. The capacitor must absorb a 100Hz current of amplitude P/V_o = 10A, giving:

$$V_{ripple,pk} = \frac{P}{2\pi(2f_0)C_{out}\,V_o} = \frac{4000}{2\pi \cdot 100 \cdot 3400\,\mu \cdot 400} = 4.68\,V\ (1.17\ \%) \tag{11}$$

Five 680 µF / 450 V electrolytic capacitors in parallel (C_{out} = 3400 µF, the same capacitor type as in [1]) were selected. They cover the hold-up requirement with a 59 % margin and keep the

DC ripple below 1.2 % of the bus voltage, which is the reason why the form factor is practically one. The 100 Hz RMS current of the capacitor bank is $P/(\sqrt{2}\cdot V_o) = 7.07$ A (1.4 A per capacitor).

The high-frequency RMS current of the capacitor is reduced by interleaving. Using the normalised expressions of [2], $I_{C,HF} = I_{in}\cdot\sqrt{[D(1-D)]}$ for a single stage and $I_{C,HF} = (I_{in}/2)\cdot\sqrt{[(1-2D)-(1-2D)^2]}$ for two phases ($D \le 0.5$), averaging over the line cycle at 4 kW gives 7.67 A for a single stage and 4.00 A for the interleaved stage, a 48 % reduction (CCM approximation, Fig. 3).

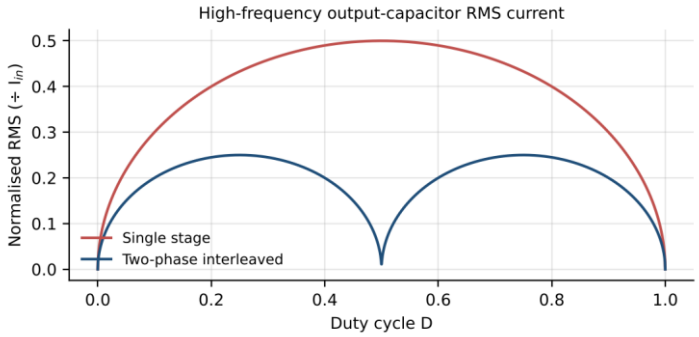


Figure 3. Normalised high-frequency RMS current of the output capacitor

3.6 Input filter and source inductance

The input network between the 700 µH source and the bridge has two branches. A 2.2 µF X-capacitor with a 2 Ω series damping resistor (C_x, R_d) limits the resonance with the source inductance ($f = 1/(2\pi\sqrt{(L_g C_x)}) = 4.06$ kHz, characteristic impedance $\sqrt{(L_g/C_x)} = 17.8$ Ω). A series L_f–C_f branch (47 µH and C_f) is tuned to the switching frequency and acts as a trap:

$$C_f = \frac{1}{(2\pi f_{sw})^2 L_f} = \frac{1}{(2\pi \cdot 10^5)^2 \cdot 47\,\mu} = 53.9\,nF \qquad (12)$$

In the interleaved converter the dominant input ripple is at $2f_{sw}$, but the trap is also intended to remove any residual component at the fundamental switching frequency that appears when the two phases are not perfectly identical (for example due to component tolerances in practice). The filter is intentionally small because the interleaving has already cancelled a large part of the ripple. Because the input current is in phase with the source voltage, the voltage drop across L_g is in quadrature with it. Its magnitude is $\omega L_g I_{in,rms} = 314.2 \cdot 700\,\mu \cdot 17.39 = 3.82$ V (1.66 %), so the fundamental of the PCC voltage is 229.97 V, i.e. practically unchanged. The PCC voltage THD is then determined only by the harmonic content of the current multiplied by ωL_g at the harmonic frequency.

3.7 Semiconductor stresses

Table 2. Approximate device stresses (ideal, lossless operation).

Device	Voltage stress	Current stress	Comment
Bridge diodes D1–D4	325 V (peak line)	7.83A avg, 12.3A rms	Iavg = √2·Iin,rms/π
Switches FET1, FET2	400 V	≈ 13.5A peak, 8.7A rms (ind.)	Shifted by 180°
Boost diodes D5, D6	400 V	5.0 A average each	Io/2 per phase
Output capacitors	400 V	7.07A (100Hz) + 4.0A (HF) RMS	5 × 680 µF

In the PLECS model the diodes and switches are ideal (V_f = 0, R_{on} = 0). In a real implementation 650 V devices would be used, for example SiC Schottky boost diodes as in [1] and fast 650 V switches; the numbers above give the current ratings.

3.8 Summary of power-stage parameters

Table 3. Power-stage parameters used in the PLECS model.

Parameter	Value	Origin
Line voltage / frequency	230 V RMS / 50 Hz	Specification
Source inductance Lg	700 µH	Specification
DC bus voltage Vo	400 V	> Vpk = 325 V
Output power / load	4000 W / 40 Ω	Specification, Vo^2/P
Switching frequency	100 kHz	Compact magnetics, ripple at 200 kHz
Boost inductors Lh (2×)	250 µH	Eq. (7), 4 A max ripple
Output capacitance Cout	3400 µF (5 × 680 µF)	Hold-up ≥ 2133 µF, ripple 1.2 %
X-capacitor / damping	2.2 µF / 2 Ω	Damping of Lg–Cx resonance
Trap La / Ca	47 µH / 53.9 nF	Tuned to 100 kHz
Carrier phase shift	180° (5 µs delay)	Interleaving

4. Control Design

The control follows the structure of the reference design [1] The diagram of Fig. 4 shows the implementation in the PLECS model.

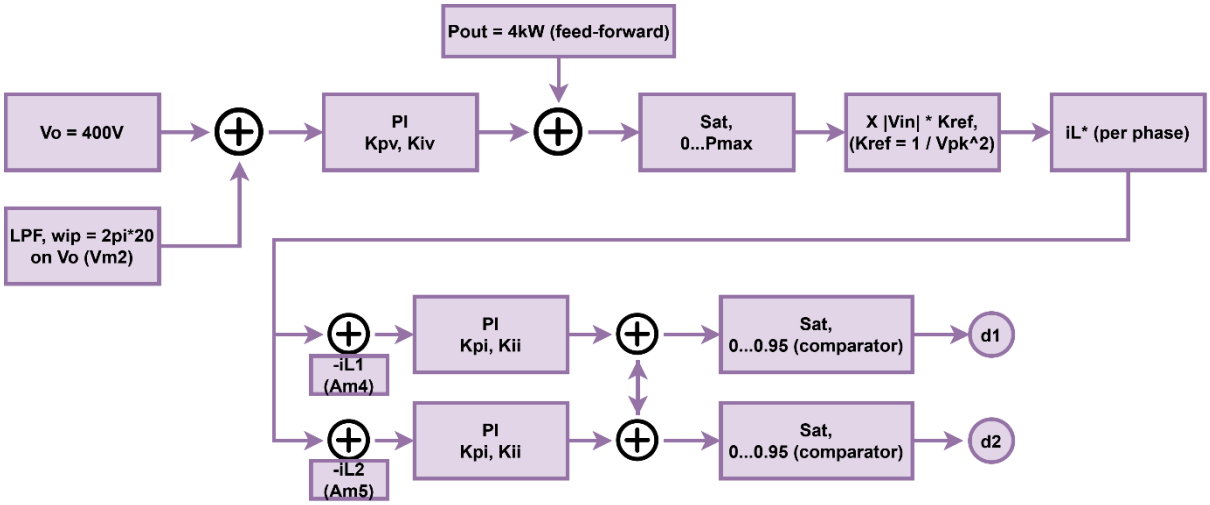


Figure 4. Block diagram of the implemented control.

4.1 Voltage loop (power reference)

The outer loop regulates the bus voltage and produces a power reference P_{ref}. For a lossless converter the capacitor energy obeys

$$C_{out}\, V_o\, \frac{dv_o}{dt} = P_{in} - P_{out} \tag{13}$$

so the plant from power to voltage is an integrator, $G_v(s) = 1/(C_{out}V_o s)$, with an additional damping term $2P/V_o$ when the load is a resistor. A PI controller sized for a crossover frequency f_{cv} = 3Hz is

$$K_{pv} = 2\pi\, f_{cv}\, C_{out}\, V_o = 25.6\, W/V, \quad K_{iv} = K_{pv}\, \frac{\omega_{cv}}{5} = 96.6\, W/(V \cdot s) \tag{14}$$

The integral zero is placed one fifth of the crossover frequency (0.6 Hz), which gives a phase margin of about 70° for the ideal integrator plant. The measurement of V_o passes through a first-order low-pass filter $\omega_{lp} = 2\pi \cdot 20$ rad/s. This is essential: the bus voltage contains 4.68 V of 100 Hz ripple, and if it reached the PI controller unattenuated it would modulate the power reference and put a third harmonic into the line current. The filter attenuates the 100 Hz ripple by a factor $1/\sqrt{(1+(100/20)^2)} = 0.196$ and, together with the low crossover frequency, keeps the modulation of the reference at about 23 W (0.6 % of 4 kW).

The controller output is added to a feed-forward of the rated power $P_{out} = 4$ kW, so that the integrator only has to correct small errors, and the sum is limited to 0 … $P_{max} = 4.5$ kW, which is also the overload (current-limit) protection (13.8 A per phase). With the damping term included, the actual loop crossover is 2.0 Hz (Fig. 5); the design target is a compromise between the speed of the response and the 100 Hz ripple injection.

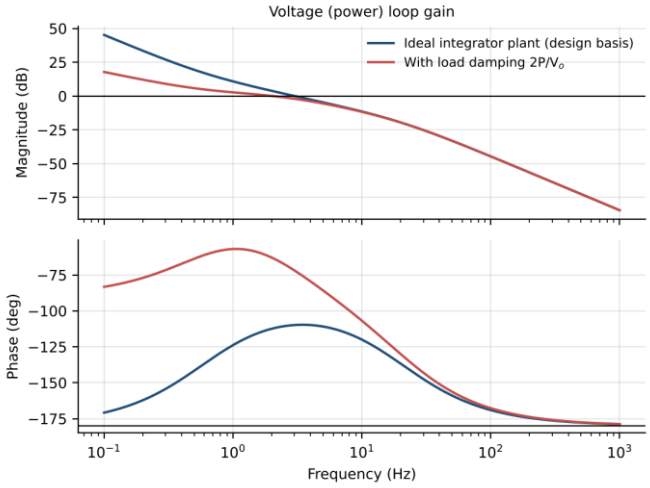


Figure 5. Voltage-loop gain L(s) = C(s)·LPF(s)·Gv(s): crossover 3.0 Hz and 70° phase margin for the ideal integrator plant (red: with load damping, crossover 2.0 Hz).

4.2 Current reference generation

The power reference is converted into an inductor-current reference that follows the shape of the rectified line voltage. The reference of each phase is

$$i_L*(t) = P_{ref}\,|v_{in}(t)|\,K_{ref}, \qquad K_{ref} = \frac{1}{{V_{pk}}^2} = 9.45 \times 10^{-6}\,V^{-2} \tag{15}$$

This choice normalises the reference so that P_{ref} is exactly the power drawn from the line. With $v_{in} = V_{pk}|\sin \omega t|$ the total input current is $2 \cdot i_L*$ and the average input power is

$$\frac{1}{T}\int v_{in} \cdot 2\, i_L* \, dt = \frac{2\,P_{ref}}{{V_{pk}}^2} \cdot \frac{{V_{pk}}^2}{2} = P_{ref} \tag{16}$$

The same reference is applied to both phases, which makes the two phase currents equal. A limitation of this fixed K_{ref} is that it is tuned for $V_{pk} = 325$ V; for other line voltages the reference would be divided by the measured amplitude, as the $1/V_{avg}^2$ term of Fig. 6 in [1].

4.3 Current loops and duty-cycle feed-forward

Each phase is an independent boost converter whose inductor current obeys $L_b\, di/dt = v_{in} - (1 - d)V_o$. Writing the instantaneous duty as the feed-forward value that would balance the volt-seconds plus a correction,

$$d = d_{ff} + d_c, \quad d_{ff} = 1 - \frac{|v_{in}|}{V_o} \tag{17}$$

the small-signal duty-to-current transfer function is $G_i(s) = V_o/(sL_b)$. The feed-forward removes the large line-voltage disturbance from the loop, so the PI controller only handles the remaining error. A PI controller with crossover f_{ci} = 10 kHz (one tenth of the switching frequency) and its zero at $\omega_{ci}/10$ is

$$K_{pi} = \frac{2\pi f_{ci} L_b}{V_o} = 0.0393, \quad K_{ii} = K_{pi} \frac{\omega_{ci}}{10} = 247 \tag{18}$$

The duty cycle is limited to 0-0.95 and compared with a 100 kHz triangular carrier. The second phase uses the same carrier delayed by half a period ($T_{del} = 1/(2f_{sw})$ = 5 µs), producing the 180° interleaving. The PWM delay of about $T_{sw}/2$ reduces the phase margin from 84° to 66° (Fig. 6), which is still comfortable. Because both loops measure their own inductor current (Am4 and Am5) and track the same reference, no additional current-balancing loop is needed in the ideal simulation; in hardware, with component tolerances, each loop would still force its own phase current towards the common reference, which is the role of the balancing loop in [1].

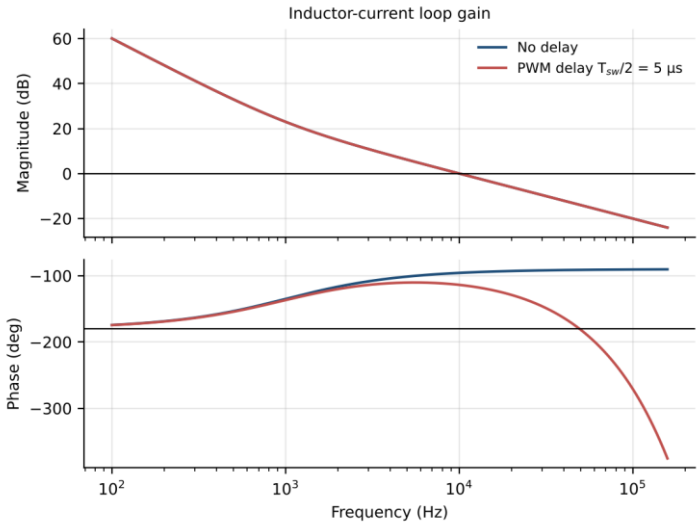


Figure 6. Current-loop gain.

Table 4. Controller parameters.

Controller parameter	Value	Meaning
Kpv, Kiv	25.6, 96.6	Voltage PI, 3 Hz, zero at 0.6 Hz
ωlp	2π·20 rad/s	Low-pass on Vo (100 Hz ripple 0.196×)
Pmax	4500 W	Power/current limit
Kpi, Kii	0.0393, 247	Current PI, 10 kHz, zero at 1 kHz
Kref	9.45×10^-6 V^-2	1/Vpk²
Duty limits	0 … 0.95	Minimum off-time
Carrier	100 kHz, 0° / 180°	Interleaving

5. PLECS Model

The complete model (Fig. 7) contains the power circuit at the centre, the two current-control chains below and above it, the voltage loop on the right and the measurement blocks on the left and right. The simulation runs for 0.5 s with the automatic solver and a maximum step of 1 µs.

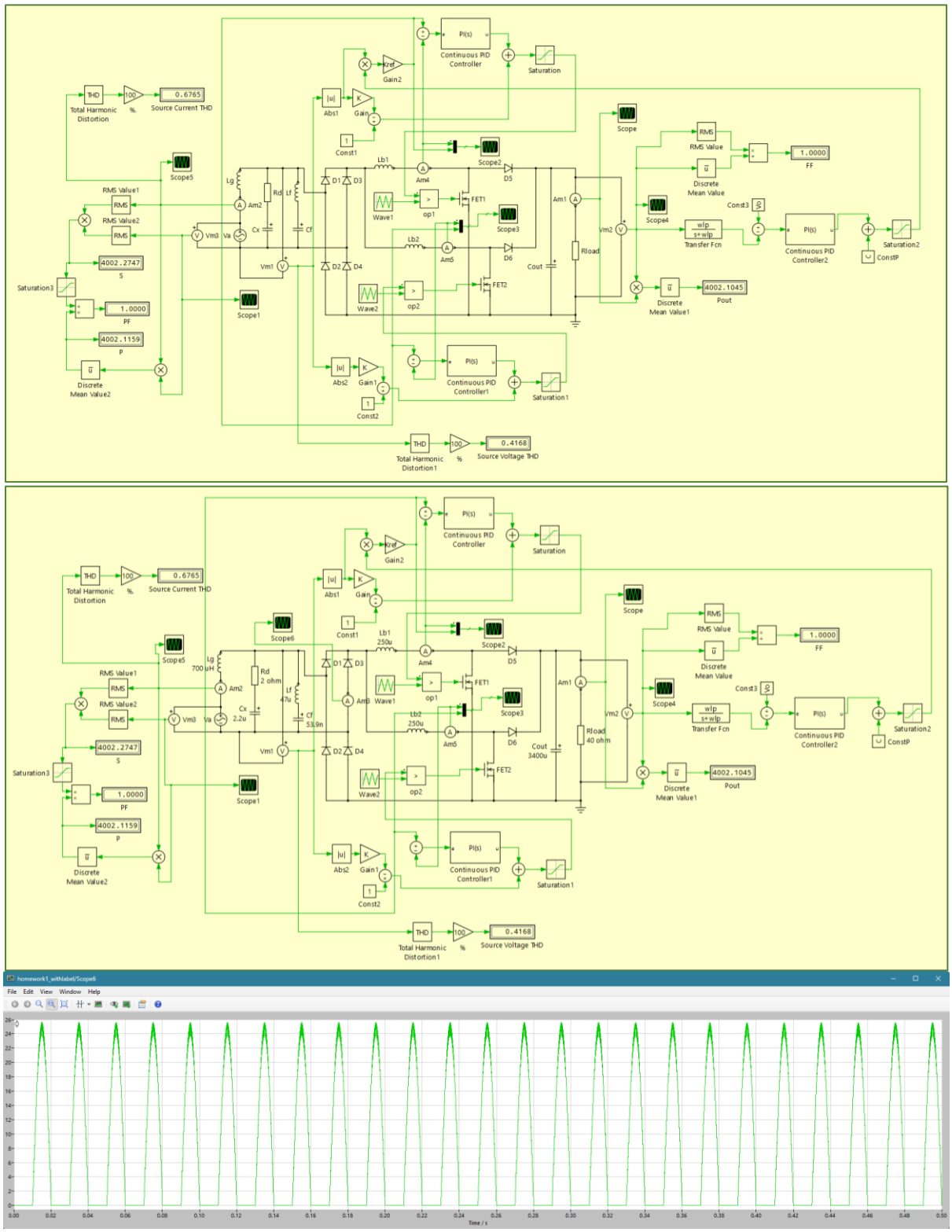


Figure 7. PLECS schematic of the interleaved boost PFC and its controller, complete schematic representation with calculated values, and rectifier current

- **Source and PCC:** AC source V_a (325.3 V, 50 Hz), L_g = 700 µH, voltmeter Vm1 at the PCC and Vm3 at the source, ammeter Am2 for the source current.
- **Input filter and bridge:** C_x with R_d, L_f–C_f trap, diodes D1–D4.

- **Boost stages:** L_{b1} / FET1 / D5 and L_{b2} / FET2 / D6, 250 µH each, with ammeters Am4 and Am5; C_{out} = 3400 µF, initial voltage 400 V, R_{load} = 40 Ω; Am1 and Vm2 measure the output current and voltage.
- **Control:** three continuous PI blocks (voltage, current 1, current 2), the low-pass transfer function with initial state V_o/ω_{lp}, the K_{ref} gain and the $|v_{in}|$ absolute-value blocks, saturations and two triangular carriers (second carrier delayed by T_{del}).
- **Measurements:** RMS blocks, discrete mean blocks (window of 2000 samples × 10 µs = 20 ms, one line cycle), THD blocks (fundamental 50 Hz), and displays for the apparent power S, the real power P, the power factor, the output power, and the form factor ($V_{o,rms}/V_{o,avg}$).

The quantities are computed as

$$PF = \frac{P}{S} = \frac{1/T \int v\, i\, dt}{V_{rms}\, I_{rms}}, \qquad THD = \frac{\sqrt{\Sigma\, I_h^{\,2}}}{I_1}, \qquad FF = \frac{V_{o,rms}}{V_{o,avg}} \tag{19}$$

6. Simulation Results

At the end of the 0.5 s simulation, the displays of the PLECS model give the values of Table 5. The measurement blocks average over one line cycle (20 ms).

Table 5. Simulation results versus specifications.

Quantity	Requirement	PLECS result	Status	Note
Power factor (P/S)	> 0.99	0.99996 (display 1.0000)	Met	P = 4002.12 W, S = 4002.27 VA
Source current THD	< 3 %	0.6765 %	Met	'Source Current THD'
PCC voltage THD	< 2 %	0.4168 %	Met	Measured at Vm1
Form factor of Vo	≈ 1	1.0000	Met	Ripple 1.2 % pk
Output power	4000 W	4002.1 W	+0.05 %	Mean of vo·io
Source RMS current	17.39 A	17.40 A	Agrees	S/Vrms

- The power factor is essentially unity. The source RMS current deduced from the displays, S/230 = 17.40 A, matches the analytical 17.39 A, and the input power (4002.1 W) equals the output power (4002.1 W) because the semiconductors are ideal.
- The source-current THD is 0.68 %, about one fifth of the limit. Because the PCC voltage drop is $\omega L_g \cdot I$ and the current is almost pure sine, the PCC voltage THD is only 0.42 %, one fifth of its 2 % limit.
- The form factor is 1.0000 because the 100Hz ripple is only 4.7 V peak on 400 V: FF ≈ 1 + $(V_r/V_o)^2/4$ ≈ 1.00003.
- The output power is 0.05 % above 4kW, the power loop is accurate to well within 1 %.

The current distortion is expected to come mainly from two sources. The first is the 100Hz ripple that is still present in the power reference through Kpv: with a 4.68 V ripple, the 0.196 low-pass attenuation and Kpv = 25.6, the reference is modulated by about 23 W (0.59 %), which creates a third-harmonic component of the order of half that value, i.e. about 0.3 % (an estimate). The second is the short discontinuous-conduction interval at the zero crossings together with the residual switching ripple. For an actual application, a 50A Schottky diode (at least) can be used to provide a safety margin, based on the rectifier arm current (approximately 25.7A).

6.1 Waveforms

The following figures show the waveforms recorded with the scopes of the model.

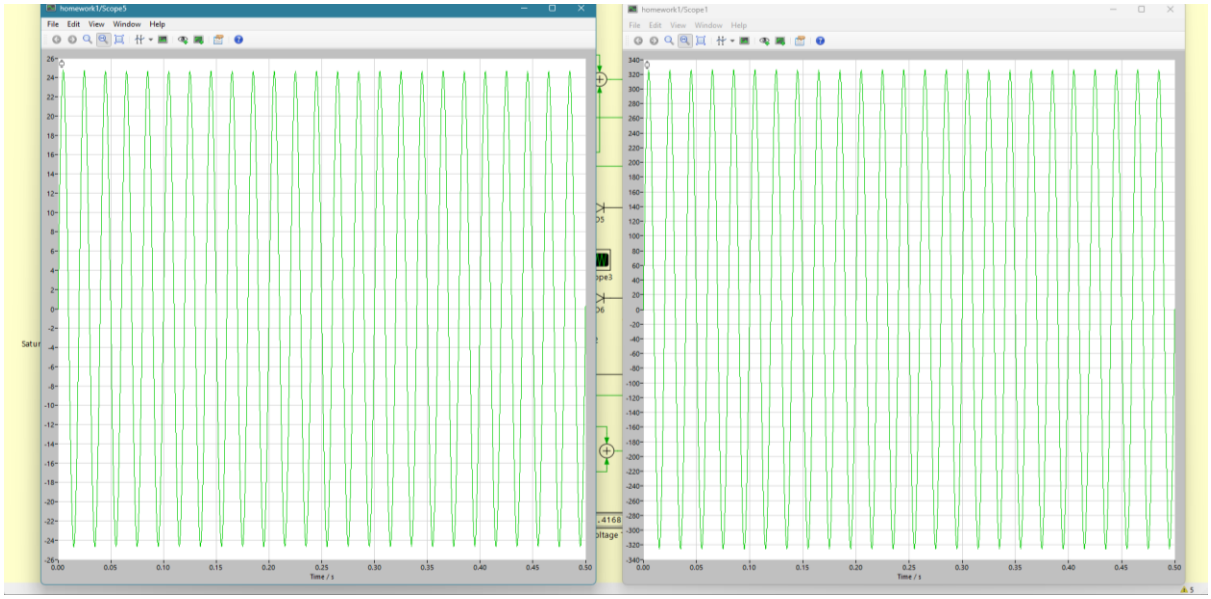


Figure 8. Source voltage and source current in steady state.

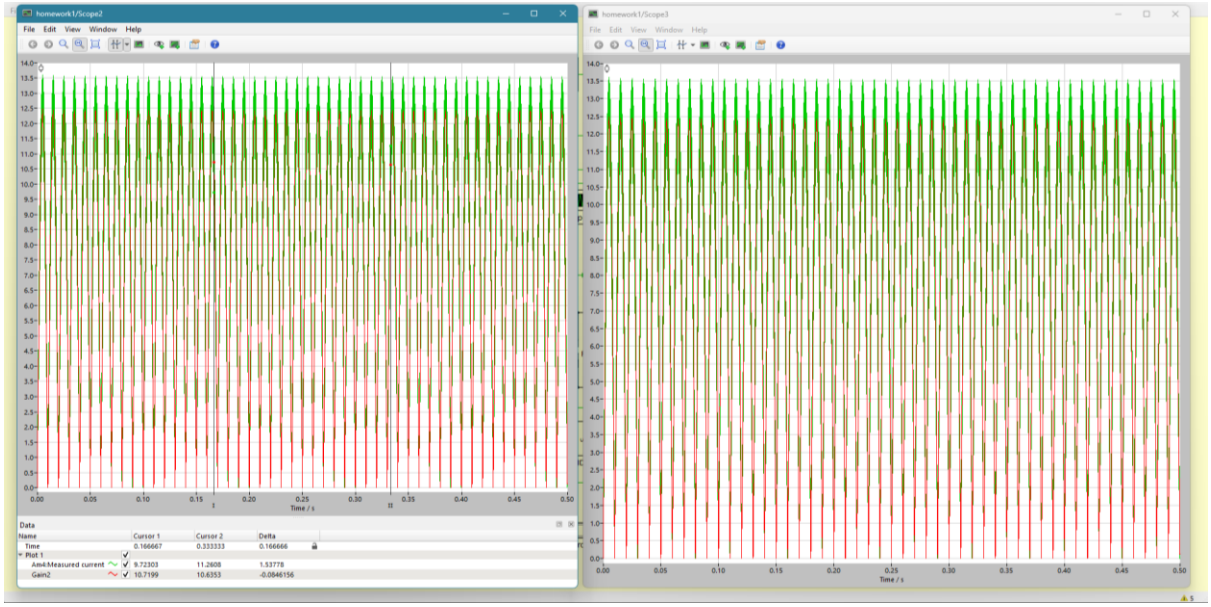


Figure 9. Interleaved inductor currents and their sum at the peak of the line Voltage.

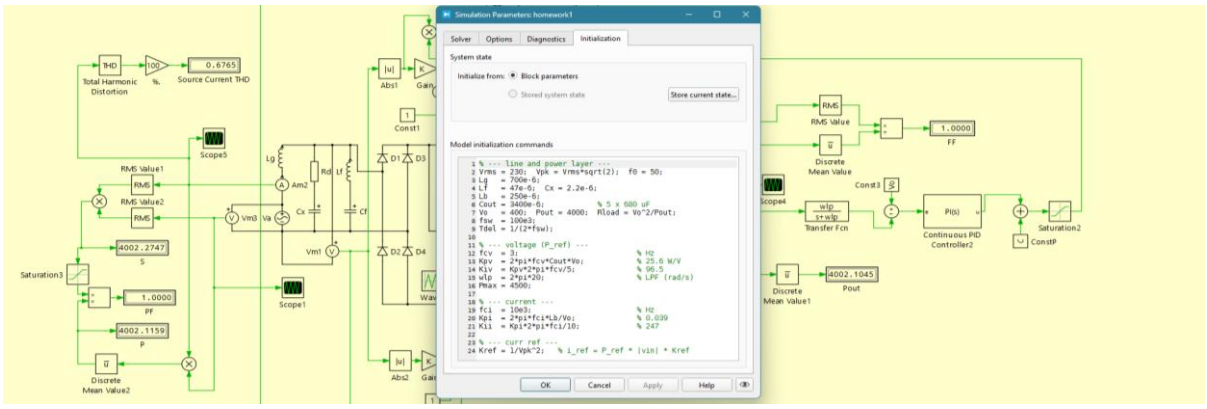


Figure 10. Settings and simulation script.

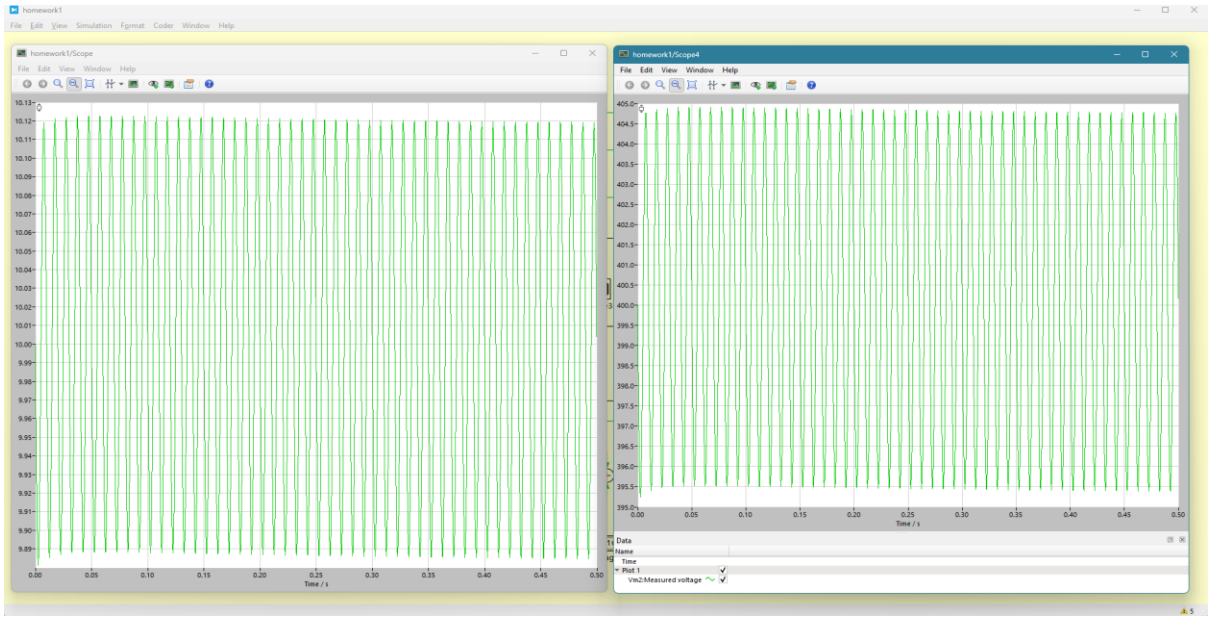


Figure 11. DC output voltage and output current.

Expected behaviour, according to the analysis: at the line peak each inductor shows a ripple of about 2.4 A peak-to-peak and the sum shows 1.9 A; around v_{in} = 200 V and v_{in} = 300 V the ripple of the sum is minimum and 2.0 A respectively; and the bus voltage ripples by about ±4.7 V at 100 Hz.



Figure 12. Fourier of input and output currents.

7. Comparison with the Reference Designs

Table 6. Comparison with the reference designs.

Item	ROHM 3.6 kW [1]	TI 350 W [2]	This design
Output power	3.6 kW	350 W	4 kW
Input voltage	180–265 V AC	85–265 V AC	230 V AC
DC bus	390 V	385 V	400 V
Phases	2	2	2

Switching frequency	65 kHz	100 kHz	100 kHz
Boost inductor	680 µH	600 µH	250 µH
Output capacitance	4 × 680 µF	220 µF	5 × 680 µF
Control	dsPIC: V loop, I loop, balancing	UCC28528 + UCC28220	V (power) loop, 2 × I loops
PF / THD	0.9965 / ≈ 3 % (full load, 230 V)	—	0.99996 / 0.68 % (ideal)

The structure is the same as in [1], the voltage loop is slower than the current loop by a large factor, and the inductor-ripple, capacitor and filter calculations follow [1] and [2]. The inductance is smaller than in the ROHM board because the accepted ripple is larger (33 % of the phase peak) and because the cancellation between phases reduces its effect at the input. The better THD and PF values of the simulation compared with the hardware board come from the ideal switches, no dead time, no current-sensing error and no losses; they show the achievable limit of the control strategy rather than a hardware prediction.

8. Conclusion

A two-phase interleaved boost PFC with an output of 400 V was designed to meet the high-power-quality requirements for a 4 kW single-phase rectifier supplied from 230 V through a 700 µH source inductance. The power stage was dimensioned analytically (2 × 250 µH, 3400 µF, 100 kHz, damped LC input filter), the control was built as a cascade of a power-reference voltage loop (3 Hz) and two current loops (10 kHz) with line-voltage feed-forward, and the whole system was verified in PLECS. All specifications are suitable: PF ≈ 1, source-current THD of 0.68 % (<3 %), PCC voltage THD of 0.42 % (<2 %), and a form factor of 1.0000.

Limitations of the study is assumed like: The semiconductors are ideal, so losses, dead time and reverse recovery are not included; the reference gain Kref assumes a fixed line amplitude; the controller is implemented in continuous time, and the discrete-time delay of a digital implementation would require a re-check of the phase margins; and the ripple and capacitor-current figures use CCM approximations. But of course, this system must be corrected with real life conditions. If we had used a standard boost converter, we could have opted for a larger core, thicker Litz wire, and a design that would be more difficult to manufacture; however, I believe the current design makes things easier.

References

[1] ROHM Semiconductor, “3.6 kW Interleaved Boost PFC Reference Design (REF67010-IPFC_3.6kW-EVK-A04),” Application Note No. 67AN069E Rev. 001, Nov. 2024.

[2] M. O’Loughlin, “An Interleaving PFC Pre-Regulator for High-Power Converters,” Texas Instruments, Topic 5, SLUA746.

[3] B. Tamyurek, Power Electronics I – Assignment 1, Gazi University, Department of Electrical Engineering, 24 Oct. 2026.

**The other graphs are Python outputs.*

Appendix A. PLECS Initialisation Script

```
% --- line and power layer ---
Vrms = 230;  Vpk = Vrms*sqrt(2);  f0 = 50;
Lg   = 700e-6;
Lf   = 47e-6;  Cx = 2.2e-6;
Lb   = 250e-6;
Cout = 3400e-6;          % 5 x 680 uF
Vo   = 400;  Pout = 4000;  Rload = Vo^2/Pout;
fsw  = 100e3;
Tdel = 1/(2*fsw);

% --- voltage (P_ref) ---
fcv  = 3;                    % Hz
Kpv  = 2*pi*fcv*Cout*Vo;          % 25.6 W/V
Kiv  = Kpv*2*pi*fcv/5;            % 96.6
wlp  = 2*pi*20;                % LPF (rad/s)
Pmax = 4500;

% --- current ---
fci  = 10e3;                 % Hz
Kpi  = 2*pi*fci*Lb/Vo;            % 0.039
Kii  = Kpi*2*pi*fci/10;           % 247

% --- curr ref ---
Kref = 1/Vpk^2;   % i_ref = P_ref * |vin| * Kref
```